

Grandperret Alexi

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[LinkedIn Profile](#)

Aerospace Engineering Student, specializing in propulsion and industrial production.

Seeking a 6-month capstone engineering internship starting February 2027.

EDUCATION

Institut Polytechnique des Sciences Avancées (IPSA)

Toulouse, FR

Fluid Mechanics (turbomachinery, thermodynamics, turbulence, gas dynamics).

Sept. 2022 – Sept. 2027

Propulsion (turboprop, electric, nuclear, rocket). *Structures* (linear elasticity, finite element analysis). *LEAN* manufacturing, quality management, aerospace law, accounting.

University of Pisa

Pisa, IT

Aerospace dynamic systems, spacecraft technology, electric propulsion, rocket propulsion, space mechanisms and structures.

Aug. 2025 – Feb. 2026

EXPERIENCE

Assistant Director and Project Manager Intern (GRUAU)

June 2026 – Sept. 2026

Gruau Argentan Site

Orne, FR

- Led refrigerant transition project: conducted **thermal, thermodynamic**, and **CFD** analysis for a refrigerated enclosure system
- Participated in executive meetings, supported site workforce management, tracked weekly **site production** metrics and financial performance.
- **Conducted business development outreach** to multiple clients, successfully securing agreements and retaining securing a new client agreements for Gruau.
- Implemented 5S methodology, contributed to **ISO 9001** certification renewal process, and established manufacturing procedures to ensure production continuity.

Production and Process Intern (Strate Composites)

June 2024 – July 2024

Strate Argentan Site

Orne, FR

- Designed and deployed a **vacuum gripping system** and transport carts to improve material handling and production flow efficiency
- Authored manufacturing procedures and technical documentation to standardize and sustain composite parts production processes
- **Conducted on-site client interventions** to determine error responsibility and **negotiate product deliveries and returns**

Quality Control (Contract) (Magneti Marelli)

June 2023 – July 2023

Marelli Argentan Site

Orne, FR

- **Supervised quality control operations:** received defective parts, performed dimensional measurements on CMM, conducted **vacuum testing**, oscilloscope measurements, and extreme temperature condition testing.
- Led problem resolution analyses (5D/8D) using **LEAN tools** (5 Why, Ishikawa, Gemba Walk, Kaizen, 5S, Kanban, FIFO).
- Collaborated closely with production and engineering teams, as well as clients, to ensure quality standards and drive continuous process improvement.

PROJECTS

Personal Project, Rocket Digital Twin | *Matlab, Simulink*

2026 – Present

- Rocket digital twin: thermodynamic and turbopropulsive analysis of a liquid-propellant rocket engine (cryogenic / turbopump-driven), flight simulation, and trajectory optimization.

Aerospike Nozzle Design | *Fluid Mechanics*

2026

- Thermodynamic and aerodynamic modeling of an aerospike nozzle using the method of characteristics (Riemann invariants), mapping Mach/pressure/temperature gradients.

Fred Et Rick Space | *CATIA V5, <https://fredetrickspace.wixsite.com/aegis>*

2025

- Designed a crewed spacecraft for 10 occupants: sized overall architecture and coordinated onboard systems (life support, power, thermal management).

SOFT SKILLS AND TECHNICAL COMPETENCIES

Languages: French (native), English (native), Spanish (fluent), Italian (B1), Portuguese (A2)

Software: Python, Matlab, Simulink, Nastran, Patran, LaTeX, CATIA V5, SolidWorks, STAR-CCM+

Activities: Music: piano (14 years), violin (3 years), orchestra and chamber music participation. Sports: tennis (6 years), soccer (12 years), kayaking (3 years), badminton (3 years). Travel: visited 9 countries, spent junior year of high school in the United States.

Soft Skills: open-mindedness, adaptability, teamwork, curiosity, strong interpersonal skills, rigor.